THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): An electromagnet loses its magnetism when the current is switched off. Reason (R): Soft iron does not retain magnetism. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Two nichrome wires A and B are of equal length; A is thicker. The same current flows in both. Which statement is correct? 1

(a) B gets hotter than A **(b)** A melts first
(c) both stay cold **(d)** A gets hotter than B

3. A coil of wire is wound on a plastic tube and then on an iron rod, with the same current. The magnetic effect is stronger with: 1

(a) the iron rod **(b)** the plastic tube
(c) equal **(d)** neither, as there is no magnetic effect

4. Assertion (A): A fuse is connected in series with the appliance. Reason (R): A fuse must carry the whole current of the appliance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Four cells in a circuit are connected one way and the bulb glows. If one cell is reversed, the bulb: 1

(a) does not change **(b)** melts **(c)** glows dimmer **(d)** glows brighter

6. An electromagnet lifts a nail when connected to a 3 V cell. With a 6 V battery it lifts more nails because: 1

(a) iron becomes heavier **(b)** the current is larger
(c) the nails are lighter **(d)** the coil is longer

7. Assertion (A): A thick nichrome wire gets less hot than a thin nichrome wire when the same current passes through it. Reason (R): A thick wire has more resistance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A student winds 50 turns on a nail and lifts 8 pins, and 100 turns and lifts 17 pins. The data show that: 1

(a) strength rises with turns **(b)** no relation
(c) strength falls with turns **(d)** pins have no mass

9. **Assertion (A):** A compass needle placed near a wire deflects when current flows through the wire. 1
Reason (R): An electric current produces a magnetic field around the wire.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A soft iron core placed inside a current-carrying coil makes a much stronger electromagnet. 1
Reason (R): Soft iron loses its magnetism easily when the current is switched off.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why a bird sitting on a high-voltage wire is not shocked. 2

12. Why do two bulbs in series glow dimly compared with a single bulb? 2

13. An electric cell goes weak. What happens to the strength of an electromagnet connected to it? 2

14. An iron draws 9 A. Choose the suitable fuse among 5 A, 10 A and 20 A and give a reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student connects a coil, a cell and a switch. She places an iron nail inside the coil and the nail picks up pins. List three changes she can make to pick up more pins. 3

16. A bulb is connected through a very long thin wire to a cell. The bulb is dim. Explain and suggest improvement. 3

17. Why do electrical appliances not work if the switch is turned off, though the plug is still in the socket? 3

OR

A bulb draws 0.3 A and a heater draws 7 A from the same supply. Which needs thicker wiring? Give a reason. 3

18. A teacher passes current through a wire wrapped in paper and a wire wrapped in iron foil near a compass. Explain the expected observations. 3

OR

Suggest three reasons why an electric bulb is not suitable as a room heater. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Dev makes an electromagnet by winding insulated copper wire around an iron nail and connecting the ends to a cell. With 50 turns it lifts 8 small iron pins; with 100 turns on the same nail and with the same cell it lifts 17 pins. When he disconnects the cell, nearly all the pins drop. 5
(i) Why is the copper wire insulated?
(ii) How does the number of turns affect the strength?
(iii) What happens to the nail's magnetism when the cell is disconnected?
(iv) Give two ways, other than changing the turns, to make this electromagnet stronger.

20. A heater draws 4.5 A, a TV 0.5 A and a bulb 0.3 A. Find the total current, say if a 5 A fuse is suitable and explain what happens if it is used. 5

OR

Explain why repair of a fan or an iron should not be done by a child, and describe the steps an electrician follows. 5

END OF QUESTION PAPER — Total: 40 Marks